VESPER INSTRUMENTS / FIELD ORB 04 / TECHNICAL DISCLOSURE

Handheld Apparatus for Spatial Scene Authoring with Independent Layer Time Control

Fictional utility patent specification | Vesper product design study

Technical field

[0001] The disclosure relates to handheld input apparatus, spatial rendering, and authoring of three-dimensional scenes. A preferred embodiment is a palm-held instrument that selects a scene element, modifies its appearance or local time, and presents the resulting edit in a chamber-scale preview.

Background

[0002] An operator working in a room-scale preview may need to alter a single moving object while other objects and environmental effects remain visible. A transport control tied to a common playhead advances or stops all tracks together. Repeated switching between the preview and an external editor can interrupt comparison of an object with its surroundings.

[0003] Spatial selection creates a second problem. Once a scene element has been acquired, slight movement of the instrument or a change in an occluding object can move the selection ray to another element. A captured target can instead retain a stable identifier across a continuous edit and across replacement of the geometry at that identifier.

Summary

[0004] A handheld controller carries independently sensed rotary inputs and a thumb actuator. A renderer associates the pose of the controller with a selection ray. The actuator can capture a selected layer or nested element under a stable identifier. The rotary inputs adjust a spatial reveal field, select a variant, or modify the local time of the captured layer. A single preview presents layers evaluated at different local times.

[0005] A held layer retains a sampled local time while other layers advance. In one embodiment the controller also jogs a held layer without releasing it into continuous playback. Variant replacement preserves the selected element’s spatial attachment and keeps the edit reversible. A protected portal can remain visible during a spatial transition to the chamber view.

Document status

Original fictional engineering based on Blade Runner 2049. The draft specifies a buildable scene-authoring analogue. It has no filing or grant number. The source concordance identifies which operations derive from the screenplay, film, or prop and which are proposed here.

Brief Description of the Drawings

[0006] FIG. 1 shows the handheld instrument 100 with three rotary input rings 110, 112, 114 and thumb actuator 120.

[0007] FIG. 2 shows a longitudinal section of the instrument and independently sensed ring positions.

[0008] FIG. 3 shows controller 100, scene host 200, spatial output 210, and the layered scene store 220.

[0009] FIG. 4 shows a calibrated chamber 230, the protected doorway volume 234, the barrier 236, and a selection ray 240.

[0010] FIG. 5 shows scene layer and nested element identifiers, a surface anchor, and a set of selectable variants.

[0011] FIG. 6 shows independent child and cake times, a hold interval, and a jog applied to the cake track.

[0012] FIG. 7 shows acquisition, editing, preview, and reversible transaction handling.

[0013] FIG. 8 shows the optional inspection terminal 280 with a separate path for reviewing an imported reference.

Terminology

[0014] A scene is a stored arrangement of digital assets and time-varying properties. A layer is an independently evaluated collection of scene elements. A scene element is an addressable object, component, light, or effect. A memory vignette is a scene authored for first-person playback. The term describes a digital representation in this specification.

[0015] A capture binds an input operation to a stable identifier for a layer or element. A hold fixes the sampled local time for one or more tracks. A jog changes a sampled local time by an operator-specified increment. An anchor is a position and orientation expressed relative to a persistent support or world reference.

Detailed Description

Instrument and sensing

[0016] In FIGS. 1 and 2, controller 100 includes a housing 102, a fixed central spine 104, a bearing sleeve 106, and rotary rings 110, 112, and 114. Each ring rides on the sleeve independently. Magnet rings and angle sensors 130 provide separate angle measurements. An inertial and optical tracking assembly 132 provides orientation and position. Processor 134 reads the controls and transmits time-stamped events to host 200. A thumb actuator 120 and haptic actuator 136 provide target capture and acknowledgment.

[0017] One prototype uses a housing approximately 150 mm long and 66 mm in maximum diameter. Each input ring has a differentiated tactile profile. The field ring 110 has broad scallops, the variant ring 112 has indexed detents, and the time ring 114 has fine knurling. A thumb paddle can be actuated without changing the working grip. Magnetic encoders may provide 12-bit angular samples at 120 Hz. The instrument can send USB or wireless packets containing a session identifier, monotonic sequence number, timestamp, pose, ring identifier, and signed angular increment.

[0018] The processor unwraps angular samples using the shortest signed difference between successive readings. It discards jumps outside a calibrated physical slew limit. A ring remains mechanically decoupled from the other rings. Firmware may also apply a small angular dead band before issuing a delta. Variant steps are quantized at detents. Field and time inputs may deliver signed continuous deltas. Alternative embodiments use optical quadrature encoders, haptic detents, or a fixed touch-sensitive annulus.

[0019] An optical front window 140 can cover a tracking illuminator or serve as an aiming reference. An optional telescoping shroud can extend around the front window while the bearing sleeve remains fixed. The production prop’s extending ring assembly motivates the appearance. Independent encoder assignments and the proposed tracking arrangement are engineering choices in this embodiment.

Host and room registration

[0020] Host 200 maintains the scene store 220, interaction state, and renderer. A calibrated transform maps controller pose to the chamber frame. A commissioning procedure registers at least three noncollinear chamber points and the position of the protected doorway. The operator may run a boresight procedure to register the aiming axis to the controller pose. The host computes ray 240 from the transformed origin and direction.

[0021] The host hit-tests visible candidates using scene geometry or a selection proxy. Before capture, it may display an outline or local light response and rank candidates by depth. A second actuation can cycle overlapping hits. On capture, the host stores the target identifier, its containing layer, and the active revision. A short haptic acknowledgment confirms capture. A capture survives changes in the geometry and in the ray until release, explicit retargeting, or removal of the identified target.

Scene State and Editing

Scene records and anchor retention

[0022] Layer records include an identifier, a visibility policy, one or more animation tracks, local clock state, and an optional parent. Element records include a stable identifier, parent identifier, anchor, local offset, asset variant index, and optional component parameters. One anchor representation uses a support identifier, triangle index, barycentric coordinates, and an orientation in the triangle tangent frame. A rigid support may instead expose a named socket.

[0023] The host derives an element pose by composing the support pose, anchor frame, and variant alignment offset. To swap an insect, the host changes the asset reference and its alignment offset while retaining the leaf anchor and target identifier. Different meshes can declare a contact frame for feet, the cake base, or a comparable interface. Component editing may modify an eye morph parameter without replacing the insect or changing the contact frame.

[0024] On a missing support or topology mismatch, the host may use a named socket or suspend the edit and signal reacquisition. It preserves the last valid pose until the operator chooses another support. A brief light sweep between assets may use two registered renders weighted by complementary masks. The stable identifier refers to the shared scene element through both renders.

Local clock evaluation

[0025] For each layer i the host stores a local epoch q(i), a host-time epoch e(i), and a playback rate r(i). At host time t it samples local time as q(i) + r(i) times (t minus e(i)). When holding a layer at tH, it sets q(i) to the previously evaluated local time, e(i) to tH, and r(i) to zero. A jog adds a signed increment to q(i), with optional frame quantization and clipping or wrap at a declared range. Releasing a hold rebases e(i) to the release time and restores the selected rate. These updates prevent a jump in pose upon release.

[0026] Each rendered frame uses a common host timestamp and evaluates each affected track against its own local time. A layer can expose independently clocked child tracks for expressions, the cake turntable, or flames. A group hold may apply to selected motion tracks while allowing independent lighting tracks to advance. Anchoring and timing are separate. An attached object can inherit the current transform of its support while its own animation remains held. A world hold can capture both the time sample and the evaluated world transform.

[0027] In a worked example, the child layer and cake layer each run at unit rate and reach local time 2.00 s. A hold captures the child layer at 2.00 s. One host second later the cake is at 3.00 s while the children remain at 2.00 s. A subsequent minus 0.25 s jog on a held cake track changes its sample to 2.75 s without altering the children. The host can undo that jog by restoring the prior cake clock state with a new epoch.

Chamber Rendering and Operation

Spatial reveal and scene handoff

[0028] Spatial output 210 may include calibrated projectors with tracked viewpoints, a see-through spatial display, or a conventional preview viewport. A physical chamber realization can use multiple projectors for surfaces and an additional tracked display for stereoscopic content. A conventional prototype uses a single real-time scene rendered from a tracked or virtual camera. Each realization uses the same authored scene data and time state.

[0029] For a segmented forest reveal, the renderer stores a chamber-space direction n and a reveal threshold d. A scene sample at x receives a visibility weight based on signed distance dot(n, x) minus d. The host can blend across a narrow band or step across ordered depth intervals. Field ring 110 moves d. A protected doorway volume 234 can override the projected content mask. A visitor event may trigger the override and an audible acknowledgment independently of the authored effect.

[0030] An edit uses a transaction record with target identifier, before state, proposed delta, and revision. During preview the renderer applies the delta to temporary state. A deliberate actuation commits a validated delta. Cancel discards it. The host checks that the target still exists and its revision matches before accepting the edit. It records accepted mutations in order and de-duplicates retransmissions using sequence identifiers. On input loss the host stops applying new deltas and discards an uncommitted edit after a timeout. Doorway protection remains under chamber control.

[0031] The preferred assignment uses field ring 110 for spatial reveal, variant ring 112 for asset or parameter changes, and time ring 114 for jog. The thumb actuator captures a target, with distinct press patterns or an additional physical key for hold and commit. Haptic patterns distinguish acquisition, a variant detent, a hold, and a conflict. Feedback may also appear on the affected scene element. No detached menu is required for the basic authoring loop.

Forest and birthday embodiments

[0032] In a forest session the operator targets an insect on a leaf, captures it, and cycles compatible insect meshes. She can retarget an eye sub-element for a continuous shape adjustment. The leaf attachment and surrounding forest persist. A directional reveal exposes the chamber in successive depth bands. A visitor signal can expose the doorway without overwriting the saved forest state.

[0033] In a birthday session the operator places a cake on a table and chooses an icing variant, sets candle states independently, and adjusts expression weights on selected faces. A sampled first-person hand animation can approach the frosting. A viewport effect applies variable edge softness while preserving the target region. The operator can hold the children while the cake turns and preview changes in flame light using a lighting track. The authored state can be exported as a versioned scene package with assets, anchors, tracks, and provenance tags.

Inspection and Implementation

Separate review path

[0034] In FIG. 8 an optional inspection terminal 280 imports an existing authored scene or a recorded spatial reference. An input station 282 can present an alignment target and collect a consent action. A review console 284 receives a read-only proxy in a review store 286. The review console can scrub the proxy and attach operator notes. It uses a distinct data path from the authoring store. Import into an editable scene, when supported, requires an explicit copy with a provenance record.

[0035] The screenplay describes an additional paired examination instrument with a chin rest, an eye-directed light, and a matching operator station. The prototype embodiment substitutes a supplied digital reference for its fictional neural source. The review console may show a recorded image stream, its annotations, and the identity of the source file. The apparatus does not infer whether a scene represents a lived event from edge blur, level of detail, or an operator judgment.

Prototype sequence

[0036] An engineer can implement the preferred embodiment with three rotary encoders on a sleeve, a tracked pose source, a microcontroller, and a host with a renderer that supports independently evaluated animation. The host loads a forest and a birthday scene. Each asset record defines a stable identifier, compatible variants, and a placement socket. The host runs capture and transaction validation before evaluating the per-layer clocks for each frame.

[0037] To verify anchor retention, attach two insect variants to the same leaf record, acquire the insect, and rotate the variant ring while moving the controller off target. Confirm that the selected identifier and leaf contact frame persist. To verify time separation, hold the child group and record its sample time for several frames while the cake track advances. Jog a separately held cake and verify that its sample changes only by the signed amount commanded. To verify the handoff, run a reveal sweep during a visitor event and confirm that the portal stays clear and the saved scene is recoverable.

[0038] Alternative embodiments include a fixed desktop controller paired with a tracked pointer, a wearable marker rather than an onboard pose sensor, wired controls through a sealed feedthrough, or a simulated ring assembly operated by touch or mouse. A virtual control surface can preserve the distinct roles of capture, reveal, variant selection, and local time. The field can also be represented by a curved boundary or a set of rooms with ordered reveal thresholds.

Reference numerals

100 controller · 102 housing · 104 spine · 106 sleeve · 110 field ring · 112 variant ring · 114 time ring · 120 thumb actuator · 130 angle sensors · 132 pose assembly · 134 processor · 136 haptic actuator · 140 window · 200 host · 210 spatial output · 220 scene store · 230 chamber · 234 protected volume · 236 barrier · 240 selection ray · 250 layer · 252 element · 254 anchor · 256 variants · 260 child time · 262 cake time · 270 transaction · 280 inspection terminal · 282 input station · 284 review console · 286 review store.

Claims

1. A spatial scene authoring system comprising a handheld controller having a pose tracker, independently sensed first and second rotary inputs, and a capture actuator, a spatial output, and a host configured to map a pose of the handheld controller to a selection ray, associate an identified layer of a scene with a captured identifier in response to actuation of the capture actuator, retain the association when the selection ray moves away from the identified layer, set a playback rate of the identified layer to zero in response to a hold command while advancing a second layer, modify a sampled local time of the identified layer in response to a signed increment from the first rotary input while the playback rate remains zero, and present the identified layer and the second layer together on the spatial output.

2. The system of claim 1, wherein the host is configured to assign the captured identifier to a nested scene element and replace an asset of the nested scene element in response to the second rotary input while retaining a surface anchor of the nested scene element.

3. The system of claim 2, wherein the surface anchor comprises a supporting mesh identifier, a triangle index, barycentric coordinates, and an orientation expressed in a tangent frame of the triangle.

4. The system of claim 2, wherein the host composes a pose of the replacement asset from a support pose, an anchor frame, and a variant-specific alignment offset.

5. The system of claim 1, wherein the host calculates the local time from a local epoch, a host-time epoch, and the playback rate and rebases the host-time epoch upon release of a hold to continue from the sampled local time.

6. The system of claim 1, wherein the handheld controller has a third independently sensed rotary input, and the host varies a chamber-space threshold in response to the third rotary input to expose portions of the chamber in depth order.

7. The system of claim 6, wherein the host applies a projection mask that excludes a calibrated doorway volume from scene projection in response to a visitor event independently of the chamber-space threshold.

8. The system of claim 1, wherein the host previews a delta in temporary state, commits the delta after verifying a target revision, and retains a before state for undo.

9. The system of claim 1, further comprising a review console configured to scrub a read-only reference in a review store separate from an editable scene store and to copy the reference into the editable scene store only in response to an explicit import operation that records provenance.

Claims Continued

10. A method of authoring a spatial scene comprising determining a target from a pose of a handheld controller, binding a stable identifier of the target to a capture state in response to a physical actuation, applying an angular input to change a scene element variant associated with the stable identifier while preserving an attachment to a supporting scene element, holding a first layer at a sampled local time, evaluating a second layer at an advancing local time while the first layer is held, and rendering the first layer and the second layer in a common spatial preview.

11. The method of claim 10, further comprising jogging the second layer by a signed rotary increment while preserving the sampled local time of the first layer.

12. The method of claim 10, wherein changing the scene element variant comprises replacing an insect mesh while preserving a contact frame on a leaf.

13. The method of claim 10, further comprising acquiring a nested eye element under a separate stable identifier and varying a shape parameter of the nested eye element while preserving the attachment of the scene element variant.

14. The method of claim 10, further comprising setting a spatially varying edge softness effect while retaining a central target region at a lower softness setting.

15. A handheld scene-authoring apparatus comprising a palm-held housing, a bearing sleeve supported by the housing, three coaxial rotary rings independently rotatable about the bearing sleeve, respective angular sensors for the three coaxial rotary rings, a pose-sensing assembly, a thumb actuator positioned for operation while holding the housing, and a processor configured to transmit time-stamped events distinguishing rotary input for spatial reveal, variant selection, and local-time jog, and to transmit a capture event responsive to the thumb actuator.

16. The apparatus of claim 15, wherein the angular sensors are magnetic sensors and the processor computes a signed unwrapped angular increment from successive angle measurements.

17. The apparatus of claim 15, wherein the rings respectively have scallops, indexed detents, and knurling that differentiate the rings by touch.

18. The apparatus of claim 15, further comprising a haptic actuator and an extendable shroud, wherein the haptic actuator distinguishes target capture from a hold acknowledgment, and the extendable shroud moves relative to the bearing sleeve.

Abstract of the Disclosure

A handheld instrument authors a spatial scene using separately sensed rotary controls and a thumb actuator. A host maps instrument pose to a selection ray and captures a stable target identifier. A captured element can change variant while retaining a spatial attachment. The host evaluates scene layers on independent local clocks and renders them together. An operator can hold one layer, jog its sampled time without resuming playback, and leave other layers advancing. A spatial reveal control changes the visible extent of a projected environment, and a protected chamber region can override the reveal. An optional review terminal presents imported references in a separate read-only path.

Drawings / Sheet 1 of 4

FIG. 1

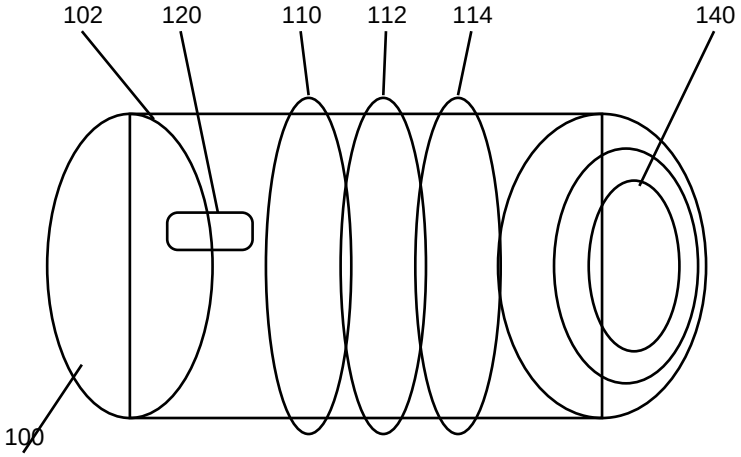
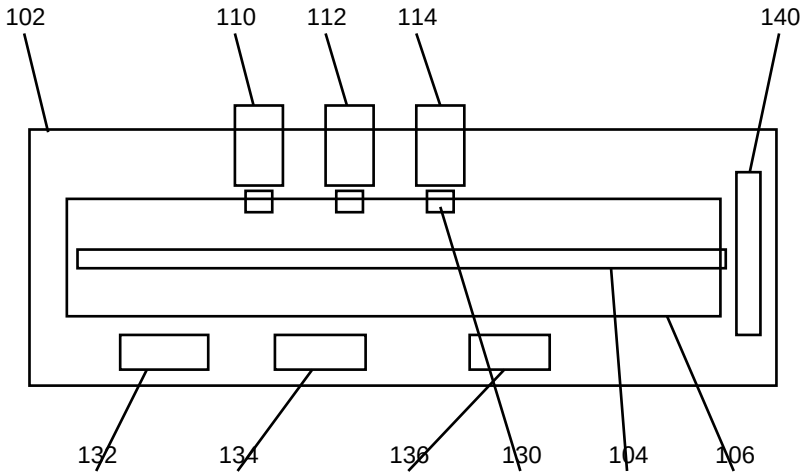


FIG. 2



Drawings / Sheet 2 of 4

FIG. 3

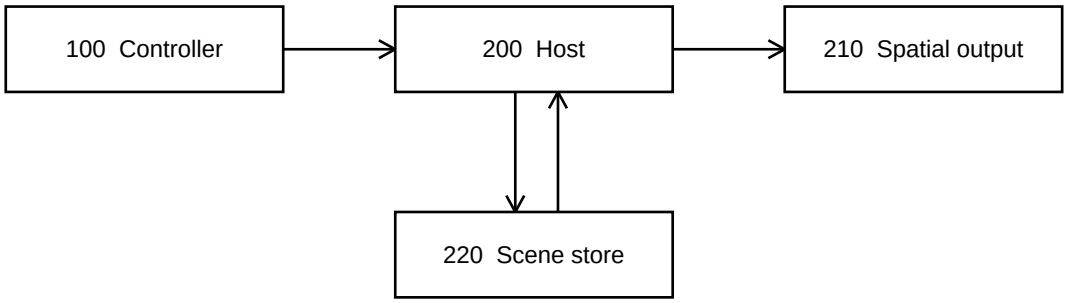
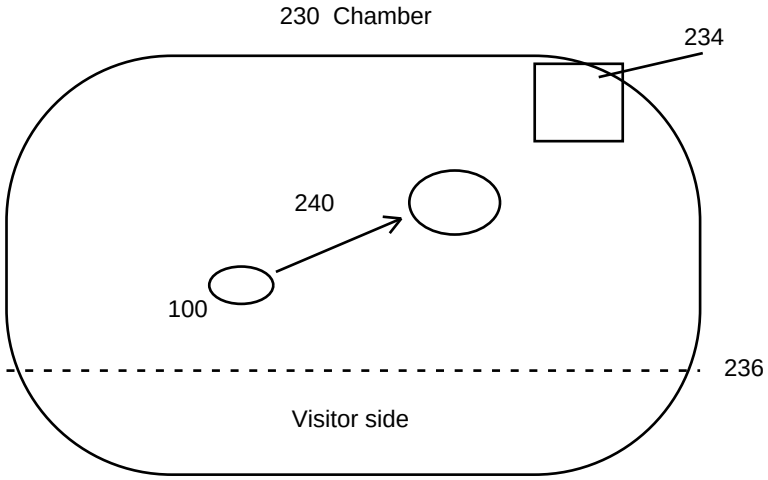


FIG. 4



Drawings / Sheet 3 of 4

FIG. 5

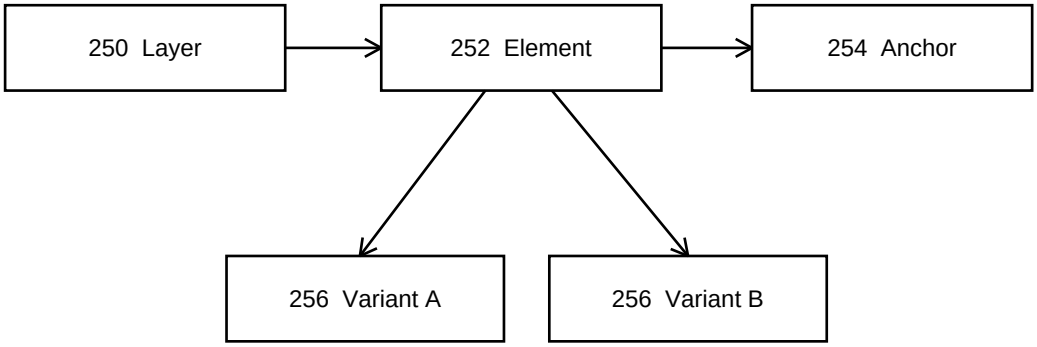
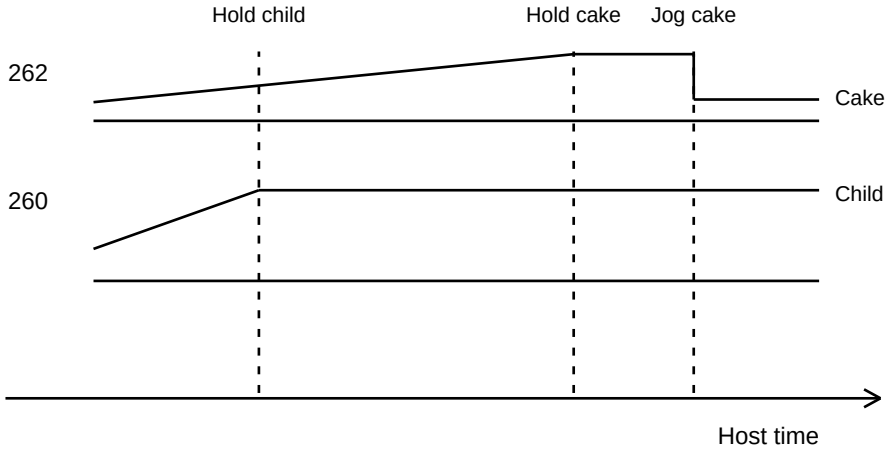


FIG. 6



Drawings / Sheet 4 of 4

FIG. 7

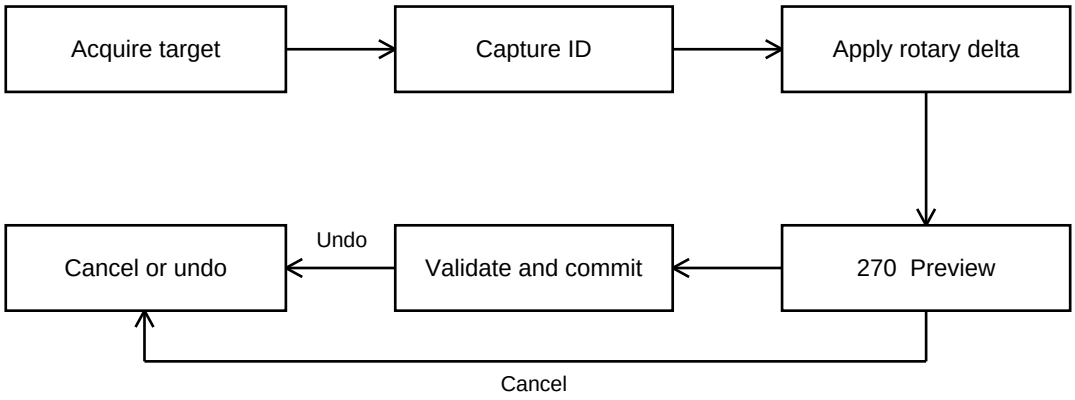
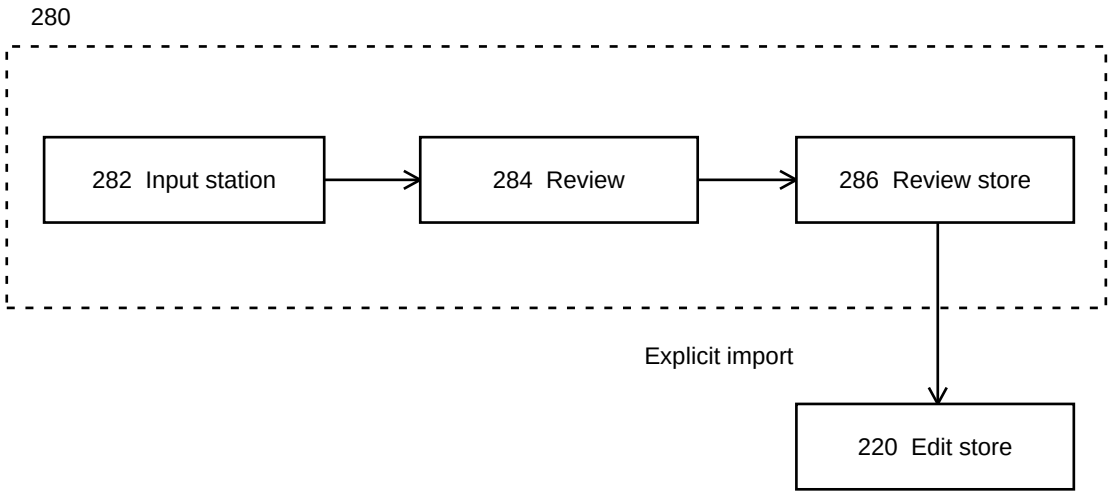


FIG. 8



Source Concordance

The film, screenplay, and prop supply different kinds of evidence. References below identify the origin of each embodiment. The claim language, Vesper brand, control assignments, and engineering details are original to this draft.

Forest and insect

Screenplay pp. 57-58. Insect substitutions take place on a leaf, followed by changes to the insect eyes. Ana manipulates images with a palm-held instrument. The door appears after a visitor alert. Stable identifiers, anchors, input assignments, and the door mask are proposed implementations.

https://assets.scriptslug.com/live/pdf/scripts/blade-runner-2049.pdf?v=1729114873

Forest and projection transitions

BUF documents the insect transitions, forest dissipation in sections, and cake variants. The threshold and crossfade models are proposed approximations for real-time rendering.

https://buf.com/films/blade-runner-2049/

Birthday authoring

Screenplay pp. 59-60. The described edits include icing, expressions, candles, softened edges, and a first-person frosting gesture. Deakins describes a finished-film effect in which children freeze while the cake turns. Independent local clocks are the proposed implementation.

https://theasc.com/article/uncanny-valley-blade-runner-2049/

Handheld prop

Propstore documents the expanding outer assembly, independently spinning inner rings, inset buttons, and labeled power switch. Mike Hill provides an animated concept. Encoder positions, tracking, and button mappings are proposed.

https://mikehill.artstation.com/projects/nXW6O

Stelline Scan

Screenplay pp. 60-62. A separate paired instrument examines recollections through an eye-directed interface. The scripted ocean comparison and wooden-horse sequence are source material for an optional review embodiment. Imported references, provenance records, and read-only review are proposed substitutes for fictional neural acquisition.

https://kinolab.org/FilmClip.php?id=1281

Additional references

Propstore Lot 131 original animatronic Memory Orb · USPTO utility application specification guidance